

SNEHA VENKATESH

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TECHNICAL SKILLS

Languages & Frameworks: Python, Java, SQL, Scala, Bash, Flask, Django, Spring Boot
Cloud & DevOps: AWS (EC2, Lambda, S3), Azure, GCP, Docker, Kubernetes, Terraform, Jenkins, GitLab CI/CD
Data Engineering & Big Data: Apache Spark, PySpark, Hadoop, Hive, Kafka, Informatica/SSIS, ETL/ELT Pipelines
Databases: Snowflake, Oracle, PostgreSQL, MySQL, MongoDB, DynamoDB, NoSQL Solutions
AI/ML & Analytics: PyTorch, TensorFlow, Scikit-learn, LangChain, OpenAI API, RAG, FAISS, Tableau, Power BI

WORK EXPERIENCE

Artificial Intelligence Python Developer | Itlize Global LLC, New Jersey, USA Aug 2025 – Present

- Executed enterprise-scale **ETL/ELT migration** from legacy **SSIS to Python-based solutions**, redesigning 50+ data workflows using **Apache Spark** and **PySpark** with optimized partitioning, reducing processing time by 60% and improving maintainability across **Snowflake data warehouse** for pharmaceutical and analytics clients.
- Designed and deployed **production-ready ML models** for predictive analytics including customer churn prediction, demand forecasting, and anomaly detection using **PyTorch**, **Scikit-learn**, and **TensorFlow**, reducing manual reporting by 40% and enabling data-driven decision-making.
- Built **GPT API-powered HR chatbot** with **RAG architecture** using **LangChain** and **FAISS**, automating employee queries on benefits, policies, and onboarding, reducing HR workload by 35% and achieving 92% user satisfaction through NLP-driven interactions.
- Implemented **Snowflake data warehouse solutions** for **Aucta Pharmaceutical** and **MMIT IQVIA**, built scalable **ETL pipelines** processing 500GB+ monthly datasets, and developed **interactive Tableau and Power BI dashboards** with predictive insights for drug trials and sales analytics.

Graduate Research Assistant | Stevens Institute of Technology, New Jersey, USA May 2025 – Present

- Designing a **multilingual benchmark** to evaluate ripple effects in large language models by analyzing editing techniques like **ROME** and **In-Context Editing** on **XLM-RoBERTa** and **BLOOM**, measuring cross-lingual consistency, logical inference, and factual knowledge preservation across diverse scripts and languages.

Software Engineer | Sigma Growth Specialists, California, USA Feb 2025 – Aug 2025

- Developed **cloud-based ETL pipelines** on **AWS** and built **AI agent systems** using **LangChain** and **GPT APIs**, integrated **Metrc** and **Blaze RESTful APIs** for cannabis compliance tracking, and implemented **Google Apps Script automation** for workflow optimization, increasing outreach efficiency by 70%.
- Served as **Technical Lead**, conducting interviews for junior developers, mentoring team members, and communicating complex technical solutions to clients, translating business requirements into scalable technical implementations.

Custom Software Engineering Analyst | Accenture, Bangalore, India Mar 2023 – Jan 2024

- Architected and deployed **Digital Service Platform** for **Star Alliance** (26 member airlines), integrating 15+ heterogeneous airline APIs using **Apigee API management** into unified **RESTful API gateway**, reducing **operational latency by 40%** and **deployment time from 6 hours to 2 hours**, supporting 500M+ annual passenger transactions and enabling seamless cross-airline digital experiences.
- Spearheaded comprehensive **CI/CD transformation** with **Jenkins**, implementing automated testing suites, security scanning, and blue-green deployment strategies on **AWS EC2** with auto-scaling, achieving **99.9% uptime SLA**, reducing manual deployment errors by 85%, and enabling rapid feature delivery across multi-region infrastructure.
- Established **enterprise DevOps standards** using **Splunk** for real-time monitoring (reducing MTTR from 30 minutes to 3 minutes), **Swagger** for API documentation, **JIRA/Confluence** for project management, and implemented **OAuth 2.0/JWT authentication** ensuring aviation security compliance and GDPR requirements across global operations.
- Led high-performing **cross-functional team of 8 engineers** across India and Hungary, conducted rigorous code reviews, mentored developers on API best practices, tested APIs with **JMeter/Postman**, and delivered critical production deployments for US, Europe, and Asia-Pacific regions with zero downtime during peak travel seasons.

Custom Software Engineering Associate | Accenture, Bangalore, India Oct 2021 – Mar 2023

- Architected robust, scalable enterprise applications using **Java**, **Spring Boot**, **RESTful APIs**, multithreading, exception handling, service orchestration, dependency injection, unit testing, concurrency control, and design patterns—achieving **95% test coverage** and **45% performance gains** through distributed system optimization.
- Engineered **big data processing pipelines** using **Apache Spark** and **Hadoop**, processing **1TB+ daily** data volumes with optimized partitioning strategies, distributed computing frameworks, and parallel processing to support enterprise-scale analytics and real-time reporting requirements.
- Designed and implemented **ETL workflows** using **Informatica PowerCenter** and **Apache Hive**, building data integration solutions with complex transformations, data quality validation, and error handling to ensure reliable data pipelines feeding downstream BI dashboards and analytical systems.

EDUCATION

Stevens Institute of Technology	Jan 2024 – May 2025
Master of Science in <i>Computer Science</i> , GPA: 3.97/4.00	Hoboken, NJ
• Coursework: Machine Learning, Artificial Intelligence, Big Data Technologies, Natural Language Processing, Web Programming, Augmented Intelligence & Generative AI, 3D Computer Vision, Algorithms, Database Management Systems, Fintech (Advanced Excel, Bloomberg Terminal)	

PROJECTS

Medical Knowledge Assistance (RAG) <i>Python, Streamlit, FAISS, Transformers, GPT-2</i>	2024
• Built Retrieval-Augmented Generation (RAG) system integrating FAISS vector database with 10M+ medical document embeddings and fine-tuned GPT-2 model, improving retrieval accuracy by 25% and deployed on AWS EC2 for 200+ healthcare practitioners.	
Product Price Optimization <i>Python, Apache Spark, GCP, Q-Learning, BERT</i>	2024
• Developed Q-learning reinforcement learning system processing 45GB Amazon reviews (5M+) using Apache Spark on GCP, integrated BERT sentiment analysis (91% accuracy), achieving 15% projected revenue increase through dynamic pricing strategies.	
FAANG AI Stock Analysis Dashboard <i>Python, Streamlit, ARIMA, LSTM, Sentiment Analysis</i>	2024
• Developed Streamlit dashboard combining real-time stock data from Yahoo Finance API, sentiment analysis of 10,000+ financial news articles, and hybrid ARIMA-LSTM forecasting to deliver AI-powered FAANG stock predictions with 2.3% RMSE.	
Multi-Cloud DevOps & AI E-Commerce Platform <i>AWS, GCP, Azure, Kubernetes, Docker</i>	2025
• Built multi-cloud e-commerce platform deployed across AWS/GCP/Azure using Kubernetes and Docker, integrated AI-powered recommendations (OpenAI, Claude AI, Amazon Bedrock), and implemented complete CI/CD pipeline with Terraform IaC.	

CERTIFICATIONS & ACHIEVEMENTS

Databricks Certified Associate Developer	2024
<i>Apache Spark 3.0</i>	
Winner - Quack Hack 2025	2025
<i>AI-powered Community Risk Monitor using Crew AI</i>	